PRECISION BIOMARKER INTELLIGENCE

AI Factory Pipeline Report

Patient: HG002	**Panel:** 160+ Analytes	**Generated:** January 31, 2026

PHASE 1	PHASE 2	PHASE 3	PHASE 4
INGESTION	**CORRELATION**	**ANALYSIS**	**INTELLIGENCE**
Blood Panel Received	Genomic Variants Mapped	7 Modules Executed	Report Generated

1. PATIENT & SPECIMEN SUMMARY

The Precision Biomarker Agent ingested a comprehensive blood panel (160+ analytes) from Patient HG002 (Genome in a Bottle reference) alongside whole-genome sequencing data processed by NVIDIA Parabricks 4.6 on DGX Spark. The canonical biomarker profile and genomic variant record were persisted for seven-module analysis.

Specimen Details

Patient ID	HG002
Report Type	Precision Biomarker Agent — HCLS AI Factory Pipeline Report
Collection Date/Time	2026-01-31 07:42
Fasting Status	Yes (12 h)
Age	44 years
Sex at Birth	Male
Ancestry	Genome in a Bottle reference
Ordering Context	Annual baseline + cardiometabolic optimization + autoimmune screen + MCED (Galleri)

Genomic Source	WGS — NVIDIA Parabricks 4.6 → GRCh38 VCF
Lab Source	Comprehensive blood panel (base + selected add-ons)

2. EXECUTIVE FINDINGS

The agent identified six priority findings ranked by clinical significance. Each finding integrates biomarker values with the patient's genomic context to produce actionable intelligence that standard reference-range interpretation would miss.

Top 6 Priority Signals

- **Prediabetic Trajectory Detected:** Fasting glucose 98 mg/dL (high-normal), HbA1c 5.5%, fasting insulin 18.6 µIU/mL (elevated), HOMA-IR 4.5 (insulin resistant). TCF7L2 rs7903146 CT heterozygous genotype confers 1.4× additional risk per T allele. Composite genetic risk score places this patient on a 5–7 year trajectory to clinical diabetes without intervention. DPP study evidence: 58% risk reduction with lifestyle modification; reversion to normal glucose reduces future diabetes risk by 56%.
- **Atherogenic Lipid Profile with APOE4 Context:** LDL-C 128 mg/dL, ApoB 108 mg/dL (elevated atherogenic particle burden), triglycerides 172 mg/dL, HDL-C 39 mg/dL (low). APOE e3/e4 genotype increases LDL receptor recycling impairment and confers dual cardiovascular + cognitive risk. Lp(a) 74 nmol/L (elevated; genetically determined, stable longitudinally). Combined risk warrants aggressive intervention.
- **MTHFR Compound Heterozygosity with Elevated Homocysteine:** MTHFR C677T CT + A1298C AC (compound heterozygous) → ~40% reduced enzyme activity. Homocysteine 13.2 µmol/L (elevated in genetic context). Methylfolate supplementation indicated; standard folic acid may be inadequately converted.
- **Hepatic Steatosis Risk (Genotype-Masked):** ALT 34 U/L and GGT 46 U/L appear "normal" by standard ranges, but PNPLA3 I148M CG heterozygous genotype (2–3× NAFLD risk per G allele) means hepatic fat accumulation occurs at lower transaminase thresholds. Agent applies genotype-adjusted ALT threshold of ≤25 U/L for this patient → flags as ELEVATED. HSD17B13 wild-type (no protective variant).
- **Subclinical Thyroid Dysfunction (DIO2 Context):** TSH 3.6 µIU/mL (high-normal) with Free T4 1.1 ng/dL. DIO2 Thr92Ala CT heterozygous genotype impairs T4-to-T3 conversion. In DIO2 carriers (~16% homozygous, ~50% heterozygous in general population), "normal" TSH may mask subclinical hypothyroid symptoms. FT3 measurement recommended.
- **Iron Storage Depletion with HFE Context:** Ferritin 22 ng/mL (below optimal), iron 72 µg/dL (low-normal). HFE H63D CG heterozygous genotype. While H63D heterozygosity has modest clinical impact, the low ferritin in a male patient warrants investigation for occult blood loss, malabsorption, or dietary insufficiency. FADS1/FADS2 variant carrier status indicates reduced omega-3 conversion efficiency → supplementation guidance adjusted.

3. BIOLOGICAL AGE ESTIMATION

The agent computes two validated biological age algorithms using the patient's lab values. PhenoAge (Levine 2018) uses 9 biomarkers plus chronological age to estimate phenotypic aging rate. GrimAge incorporates DNA-methylation surrogates for plasma proteins (GDF-15,

cystatin C, leptin, PAI-1, TIMP-1) to predict mortality and morbidity risk. GDF-15 shows the strongest individual correlation with the composite score (r = 0.664).

Biological Age Summary

Algorithm	Chronological Age	Biological Age	Delta	Interpretation
PhenoAge (Levine 2018)	44.0 years	46.8 years	+2.8 years	Accelerated aging
GrimAge (Lu 2019)	44.0 years	47.2 years	+3.2 years	Accelerated aging

PhenoAge Input Biomarkers

Biomarker	Value	Units	Contribution
Albumin	4.6	g/dL	Protective (inverse)
Creatinine	1.03	mg/dL	Neutral
Glucose (fasting)	98	mg/dL	Mild positive (aging)
C-Reactive Protein (ln)	0.88	ln(mg/L)	Positive (aging)
Lymphocyte %	32	%	Protective (inverse)
Mean Cell Volume	87.5	fL	Neutral
Red Cell Dist. Width	13.2	%	Neutral
Alkaline Phosphatase	86	U/L	Neutral
White Blood Cell Count	6.2	10^3/µL	Neutral

GrimAge Plasma Protein Surrogates

Surrogate	Measured Proxy	Value	Age Correlation	Risk Direction
GDF-15	GDF-15 (estimated)	1,180 pg/mL	r = 0.664	↑ Accelerating
Cystatin C	Cystatin C	0.92 mg/L	r = 0.52	Neutral
Leptin	Leptin (estimated)	14.8 ng/mL	r = 0.48	↑ Mild
PAI-1	PAI-1 (estimated)	38 ng/mL	r = 0.42	↑ Prothrombotic
TIMP-1	TIMP-1 (estimated)	224 ng/mL	r = 0.39	Neutral
Adrenomedullin	ADM (estimated)	44 pg/mL	r = 0.36	Neutral

Beta-2-Microglobulin	β2M	1.8 mg/L	r = 0.34	Neutral

Clinical implication: The +2.8 to +3.2 year biological age acceleration is driven primarily by the insulin-resistant metabolic profile (elevated glucose, hs-CRP, HOMA-IR) and subclinical inflammation. The prediabetic trajectory and atherogenic lipid profile are consistent accelerators. Intervention targeting insulin sensitivity, inflammation, and lipid management has the highest potential to reverse biological age acceleration.

4. GENOMIC VARIANT PROFILE

NVIDIA Parabricks 4.6 on DGX Spark processed the HG002 whole-genome sample (Genome in a Bottle reference), extracting clinically actionable variants across 15 pharmacogenomic and disease-risk loci. Variants are stored as first-class canonical objects with population frequencies and effect sizes for downstream module correlation.

Disease-Risk Variants

Gene / Locus	Variant	rsID	Genotype	Effect	Clinical Significance
MTHFR	C677T	rs1801133	CT	Het → ~35% ↓ activity	Compound het w/ A1298C
MTHFR	A1298C	rs1801131	AC	Het → ~20% ↓ activity	Compound het w/ C677T
APOE	Alleles	—	e3/e4	1 × ε4 allele	CV + cognitive risk
TCF7L2	Intron 3	rs7903146	CT	1.4× T2D risk / T allele	Prediabetic trajectory
PCSK9	—	—	Wild-type	No LOF / GOF	Standard LDL metabolism
PNPLA3	I148M	rs738409	CG	Het → 2–3× NAFLD risk	Genotype-adjusted ALT
TM6SF2	E167K	rs58542926	CC	Wild-type	No hepatic impact
HSD17B13	—	rs72613567	Wild-type	No protective variant	No NAFLD/MASH protection
HFE	C282Y	rs1800562	GG	Wild-type	No hemochromatosis risk
HFE	H63D	rs1799945	CG	Heterozygous	Mild iron loading potential
FADS1/FADS2	Cluster	rs174547	CT	Reduced desaturase	↓ Omega-3 conversion

DIO2	Thr92Ala	rs225014	CT	Het → impaired T4→T3	Thyroid context

Pharmacogenomic Variants

Gene	Star Allele	Metabolizer Status	Drug Impact	Clinical Action
CYP2D6	*1/*4	Intermediate	Codeine, tramadol, tamoxifen	Reduced activation → adjust dose
CYP2C19	*1/*2	Intermediate	Clopidogrel, PPIs, SSRIs	Reduced activation → consider alternatives
SLCO1B1	rs4149056 TC	Intermediate function	Statins (simvastatin, atorvastatin)	Myopathy risk → lower dose or rosuvastatin
VKORC1	rs9923231 AG	Intermediate sensitivity	Warfarin	Standard-to-reduced initial dose

5. CORE LABORATORY RESULTS

All 160+ analytes are presented below with standard reference intervals and flag indicators. H/L indicate values outside the reference interval. H*/L* indicate borderline or context-dependent interpretation requiring genomic correlation. The agent applies genotype-adjusted thresholds where applicable, noted in the Comment column.

Complete Blood Count (CBC) + Differential

Analyte	Result	Units	Reference Interval	Flag	Comment
WBC	6.2	10^3/µL	4.0–11.0		
RBC	5.05	10^6/µL	4.5–5.9		
Hemoglobin	14.7	g/dL	13.5–17.5		
Hematocrit	44.2	%	41–53		
MCV	87.5	fL	80–100		
MCH	29.1	pg	27–33		
MCHC	33.3	g/dL	32–36		
RDW	13.2	%	11.5–14.5		
Platelets	248	10^3/µL	150–450		
MPV	9.8	fL	7.5–12.5		

Neutrophils %	58	%	40–70		
Lymphocytes %	32	%	20–45		
Monocytes %	7	%	2–10		
Eosinophils %	2	%	0–6		
Basophils %	1	%	0–2		

Metabolic Panel (Fasting)

Analyte	Result	Units	Reference Interval	Flag	Comment
Glucose (fasting)	98	mg/dL	70–99		High-normal; TCF7L2 context → trajectory risk
HbA1c	5.5	%	4.0–5.6		Upper-normal; trend monitoring critical
Insulin (fasting)	18.6	µIU/mL	~2–20	**H***	Elevated relative to glucose → IR
HOMA-IR	4.5	index	<2.5 optimal	**H***	Insulin resistant (derived: Glu×Ins/405)
Uric Acid	6.4	mg/dL	3.4–7.0		Correlates with metabolic syndrome

Cardiometabolic / Lipid Panel

Analyte	Result	Units	Reference Interval	Flag	Comment
Total Cholesterol	212	mg/dL	<200	**H**	Borderline elevated
LDL-C	128	mg/dL	<100 optimal	**H**	APOE e3/e4 context → impaired clearance
HDL-C	39	mg/dL	≥40	**L**	Below protective threshold
Triglycerides	172	mg/dL	<150	**H**	Insulin resistance

					marker
ApoB	108	mg/dL	<90 optimal	**H***	Atherogenic particle burden elevated
Lp(a)	74	nmol/L	<75 (assay)		Borderline; genetically determined, stable
Small Dense LDL	42	mg/dL	contextual	**H***	Pattern B → atherogenic phenotype
Omega-3 Index	4.2	%	>8% optimal	**L***	FADS1/2 variant → ↓ conversion
Omega-6:Omega-3 Ratio	12.4:1	ratio	<4:1 optimal	**H***	Pro-inflammatory ratio

Liver Function & Proteins

Analyte	Result	Units	Std Reference	Flag	Genotype-Adjusted Interpretation
ALT	34	U/L	7–56		PNPLA3 CG: threshold ≤25 U/L → ELEVATED
AST	28	U/L	10–40		Within standard range
GGT	46	U/L	9–48		High-normal; correlate with TG/insulin
ALP	86	U/L	44–147		
Total Bilirubin	0.7	mg/dL	0.2–1.2		
Albumin	4.6	g/dL	3.5–5.0		Strong hepatic synthetic function
Total Protein	7.1	g/dL	6.0–8.3		
Globulin	2.5	g/dL	2.0–3.5		

Note: PNPLA3 I148M heterozygous carriers accumulate hepatic fat at ALT levels that appear “normal” by standard reference ranges. TM6SF2 E167K is wild-type in this patient. HSD17B13 is wild-type (no protective variant against NAFLD/MASH progression). The agent recommends hepatic steatosis screening (ultrasound or MRI-PDFF) despite standard ranges appearing unremarkable.

Kidney & Electrolytes

Analyte	Result	Units	Reference Interval	Flag	Comment
Creatinine	1.03	mg/dL	0.74–1.35		
BUN	16	mg/dL	7–20		
eGFR (CKD-EPI)	92	mL/min/1.73m²	≥90		Age-appropriate
Sodium	140	mmol/L	135–145		
Potassium	4.3	mmol/L	3.5–5.1		
Chloride	102	mmol/L	98–106		
CO_2 (Bicarb)	25	mmol/L	22–29		
Calcium	9.4	mg/dL	8.6–10.2		
Magnesium	2.0	mg/dL	1.7–2.2		

Thyroid Panel

Analyte	Result	Units	Reference Interval	Flag	Genotype-Adjusted Interpretation
TSH	3.6	µIU/mL	0.4–4.0		DIO2 CT: high-normal may mask symptoms
Free T4	1.1	ng/dL	0.8–1.8		Lower quartile in DIO2 carrier context
Free T3	Not ordered	—	—	—	RECOMMENDED: DIO2 carriers need FT3
TPO Ab	<9	IU/mL	Negative		No autoimmune thyroiditis
Tg Ab	<1	IU/mL	Negative		No autoimmune thyroiditis

DIO2 Thr92Ala heterozygous carriers may have impaired T4-to-T3 conversion despite “normal” TSH. The agent recommends Free T3 measurement and symptom correlation. If symptomatic, combination T4+T3 therapy may be considered.

Nutrients & Micronutrients

Analyte	Result	Units	Reference Interval	Flag	Genotype-Adjusted Interpretation
Vitamin D (25-OH)	28	ng/mL	30–100 optimal	**L***	Insufficient; supplement to >40 ng/mL
Iron	72	µg/dL	60–170		Low-normal in male
Ferritin	22	ng/mL	24–336	**L**	Below optimal; investigate cause
Homocysteine	13.2	µmol/L	5–15		MTHFR compound het → elevated risk
MMA	162	nmol/L	<370		B12 status adequate
Folate (serum)	8.2	ng/mL	>5.9		MTHFR context: methylfolate preferred
Vitamin B12	480	pg/mL	200–900		Adequate

Inflammation / Stress / Aging Markers

Analyte	Result	Units	Reference Interval	Flag	Comment
hs-CRP	2.4	mg/L	<1.0 low risk	**H***	Hepatic-origin inflammation marker
NLR (derived)	1.81	ratio	<3.0		Cellular inflammation: normal
Cortisol (AM)	15.2	µg/dL	6–18 (AM)		Within diurnal range
DHEA-S	178	µg/dL	age/sex-dependent		Age-appropriate
IGF-1	138	ng/mL	age-dependent		Mid-range for age
SHBG	31	nmol/L	10–57		Low-normal; correlates with IR
GDF-15	1,180	pg/mL	<1,200		Upper limit; aging biomarker

					(GrimAge)

Inflammatory phenotyping: hs-CRP 2.4 mg/L (hepatic-origin) with normal NLR 1.81 (cellular) suggests metabolic/hepatic inflammation rather than systemic immune activation. This pattern is consistent with insulin resistance-driven hepatic inflammation and supports the PNPLA3-context liver finding above.

Autoimmunity Screen

Analyte	Result	Units	Reference Interval	Flag	Comment
ANA Screen (IFA)	Negative	—	Negative		No signal
DNA (ds) Ab Crithidia	<1:10	titer	Negative		
Rheumatoid Factor	<10	IU/mL	<14		Negative
CCP Ab (IgG)	<20	U/mL	<20		Negative
Cardiolipin Ab (IgG)	<12	GPL	<20		
Beta-2-GP Ab (IgG)	<8	SGU	<20		
ESR	8	mm/hr	0–15		Normal
C3c	118	mg/dL	90–180		Normal complement
C4c	24	mg/dL	10–40		Normal complement

Environmental Toxins

Analyte	Result	Units	Reference Interval	Flag	Comment
Lead (whole blood)	1.2	µg/dL	<3.5 (adult)		Within range
Mercury (blood)	3.1	µg/L	<10		Within range

6. PRE-SYMPTOMATIC DISEASE TRAJECTORIES

The agent models pre-symptomatic disease trajectories by correlating biomarker trends with genetic risk scores. These trajectories identify disease progression years before clinical diagnosis, enabling early intervention during the reversible window.

Disease Trajectory Assessment

Trajectory	Risk Score	Key Drivers	Estimated Horizon	Reversibility	Priority
Type 2 Diabetes	HIGH	HOMA-IR 4.5 + TCF7L2 CT + glucose 98	5–7 years	58% (DPP)	★★★★★
Cardiovascular Disease	HIGH	ApoB 108 + Lp(a) 74 + APOE e3/e4	10–15 years	Modifiable (LDL, TG)	★★★★★
NAFLD / Steatosis	MODERATE	PNPLA3 CG + ALT 34 + TG 172 + IR	3–5 years	Highly reversible	★★★★☆
Cognitive Decline	MODERATE	APOE e3/e4 + insulin resistance	15–25 years	Lifestyle-modifiable	★★★☆☆
Autoimmune	LOW	All screens negative	N/A	N/A	★☆☆☆☆

Diabetes Trajectory Detail

The composite diabetes genetic risk score integrates: TCF7L2 rs7903146 CT (1.4× per T allele), PPARG Pro12Ala (wild-type — no protective variant), SLC30A8, KCNJ11, and GCKR. Combined with the metabolic profile (HOMA-IR 4.5, fasting glucose 98, HbA1c 5.5%, triglycerides 172), the model estimates clinical diabetes onset in 5–7 years without intervention. The Diabetes Prevention Program demonstrated 58% risk reduction with lifestyle modification, and reversion to normal glucose reduces future diabetes risk by 56%. TCF7L2 carriers specifically benefit from reduced glycemic load.

Cardiovascular Trajectory Detail

The cardiovascular trajectory combines: APOE e3/e4 (↓ LDL receptor recycling), elevated ApoB 108 mg/dL (atherogenic particle burden above 90th percentile), Lp(a) 74 nmol/L (genetically elevated, stable, confers independent risk), LDL-C 128 mg/dL, HDL-C 39 mg/dL, and triglycerides 172 mg/dL. PCSK9 is wild-type (standard response to statin therapy expected). SLCO1B1 TC intermediate function genotype indicates myopathy risk with high-dose simvastatin → rosuvastatin or pravastatin preferred.

7. GENOTYPE-ADJUSTED REFERENCE RANGES

Standard laboratory reference ranges are population averages that obscure individual genetic reality. The agent applies genotype-specific thresholds where evidence supports meaningful reclassification. The table below summarizes adjustments applied to this patient's results.

Applied Genotype Adjustments

Analyte	Standard Range	Genotype	Adjusted Threshold	Patient Value	Reclassification
ALT	7–56 U/L	PNPLA3 I148M CG	≤25 U/L	34 U/L	Normal → ELEVATED

TSH	0.4–4.0 µIU/mL	DIO2 Thr92Ala CT	≤2.5 optimal	3.6 µIU/mL	Normal → MONITOR
Homocysteine	5–15 µmol/L	MTHFR compound het	≤10 µmol/L	13.2 µmol/L	Normal → ELEVATED
Ferritin	24–336 ng/mL	HFE H63D CG	≥50 ng/mL optimal	22 ng/mL	Low → INVESTIGATE
Omega-3 Index	>8% optimal	FADS1/2 CT	>5% adjusted	4.2%	Low → SUPPLEMENT
Fasting Glucose	70–99 mg/dL	TCF7L2 CT	≤90 mg/dL	98 mg/dL	Normal → PRE-TRAJECTORY

8. PHARMACOGENOMIC GUIDANCE

The agent maps the patient's pharmacogenomic profile to CPIC (Clinical Pharmacogenetics Implementation Consortium) guidelines, generating prescribing context for commonly encountered drug classes.

Drug Interaction Profile

Drug Class	Representative Agents	Gene	Phenotype	CPIC Recommendation
Opioid Analgesics	Codeine, Tramadol	CYP2D6 *1/*4	Intermediate metabolizer	Reduced activation → may need alternative analgesic
Antiplatelet	Clopidogrel	CYP2C19 *1/*2	Intermediate metabolizer	Reduced activation → consider prasugrel or ticagrelor
Statins	Simvastatin, Atorvastatin	SLCO1B1 TC	Intermediate function	Myopathy risk → use ≤20 mg simvastatin or switch to rosuvastatin
Anticoagulants	Warfarin	VKORC1 AG	Intermediate sensitivity	Standard-to-reduced initial dose; monitor INR closely
PPIs	Omeprazole, Pantoprazole	CYP2C19 *1/*2	Intermediate metabolizer	May need dose adjustment for H. pylori eradication
SSRIs	Escitalopram, Sertraline	CYP2C19 *1/*2	Intermediate metabolizer	Standard dosing acceptable; monitor for

				increased levels
Anti-Estrogen	Tamoxifen	CYP2D6 *1/*4	Intermediate metabolizer	Reduced endoxifen formation → consider alternative if indicated
Folate Supplementation	Folic acid vs. methylfolate	MTHFR compound het	Reduced conversion	Use L-methylfolate (5-MTHF) instead of folic acid

9. MULTI-CANCER EARLY DETECTION (GALLERI)

The Galleri multi-cancer early detection test (GRAIL) uses cell-free DNA methylation patterns to screen for 50+ cancer types with a single blood draw. This is a screening test, not a diagnostic assay.

Galleri Results

Field	Result	Interpretation
Primary Result	No Cancer Signal Detected	Negative screen
Predicted Signal Origin	—	Only reported if signal detected
Follow-up Recommendation	Continue age-appropriate screening	Standard surveillance
Limitations	Does not replace standard screening	Mammography, colonoscopy, low-dose CT per guidelines

10. PERSONALIZED INTERVENTION PLAN

The agent generates an evidence-grounded intervention plan ranked by expected impact and tailored to the patient's genotype-biomarker profile. Each recommendation cites the driving finding and genetic context.

Priority Interventions

1. **Insulin Sensitization (URGENT):** Structured lifestyle modification targeting HOMA-IR reduction. TCF7L2 carriers benefit specifically from reduced glycemic load. Target: fasting glucose <90, HOMA-IR <2.5, HbA1c <5.2%. Consider metformin if lifestyle alone insufficient after 6 months.
2. **Lipid Management (URGENT):** Initiate statin therapy — rosuvastatin preferred (SLCO1B1 TC genotype contraindicates high-dose simvastatin). Target ApoB <80 mg/dL. APOE e3/e4 context increases urgency. Lp(a) 74 nmol/L is genetically fixed; manage all modifiable risk factors aggressively. Add EPA/DHA supplementation (2–4 g/day) for triglycerides and FADS1/2 variant compensation.

3. **Methylfolate Supplementation:** MTHFR compound heterozygote: switch from folic acid to L-methylfolate (1–2 mg/day). Target homocysteine <10 µmol/L. Add B6 (P5P form) and B12 as cofactors. Recheck homocysteine in 8–12 weeks.

4. **Hepatic Steatosis Screening:** Order liver ultrasound or MRI-PDFF to assess hepatic fat fraction. PNPLA3 CG genotype with insulin resistance creates high-risk context despite "normal" standard ALT. If steatosis confirmed, weight reduction of 7–10% body weight targets resolution.

5. **Thyroid Optimization:** Order Free T3 measurement. DIO2 Thr92Ala CT carrier with TSH 3.6 — if symptomatic (fatigue, cold intolerance, weight gain), consider combination T4+T3 therapy trial. TPO/Tg antibodies negative → no autoimmune component.

6. **Iron Repletion:** Investigate cause of low ferritin (22 ng/mL) in a male patient: dietary assessment, occult blood loss screen (fecal occult, celiac serologies). Supplement elemental iron with vitamin C for absorption. Target ferritin >50 ng/mL.

7. **Vitamin D Repletion:** Supplement vitamin D3 4,000–5,000 IU/day to achieve 25-OH level >40 ng/mL. Recheck in 8–12 weeks. Vitamin D sufficiency supports insulin sensitivity and immune function.

8. **Omega-3 Optimization:** FADS1/FADS2 CT variant reduces endogenous conversion of ALA to EPA/DHA. Direct EPA+DHA supplementation (2–4 g/day marine-source) recommended. Target Omega-3 Index >8%. This also addresses the triglyceride elevation and pro-inflammatory omega-6:3 ratio.

11. MONITORING SCHEDULE

Genotype-informed retest intervals replace standard annual screening where genetic context warrants accelerated surveillance.

Recommended Follow-up Timeline

Timeframe	Panel / Test	Rationale
8–12 weeks	Homocysteine, Vitamin D, Ferritin	Assess supplementation response
3 months	Fasting glucose, insulin, HOMA-IR, HbA1c	Track insulin sensitization
3 months	Lipid panel + ApoB	Assess statin response
3 months	Free T3 (new order)	DIO2 carrier thyroid assessment
6 months	Liver ultrasound / MRI-PDFF	PNPLA3 context hepatic screening
6 months	Full repeat panel (60+ analytes)	Longitudinal delta analysis
12 months	Full repeat panel + biological age	Annual trajectory comparison
12 months	Galleri repeat	Annual MCED surveillance
5 years	Lp(a) recheck	Genetically stable; infrequent monitoring
As needed	Pharmacogenomic card	Present to all prescribers

12. CANONICAL DATA PACKAGE

This is the canonical substrate for the agent: raw values + flags + provenance + derived features + longitudinal context. All downstream agents (Imaging, Genomics, Drug Discovery) reference this canonical record.

Ingestion Metadata

Canonical Record ID	PB-20260131-0742-HG002
Source	Comprehensive blood panel (base + add-ons) + Parabricks WGS
Schema Version	PB-SCHEMA-1.0
Units Policy	Single-vendor (no cross-vendor unit mixing)
Derived Features	HOMA-IR, NLR, lipid ratios, PhenoAge, GrimAge, risk bundles, trajectory models
Genomic Source	Parabricks 4.6 → GRCh38 VCF → 15 loci extracted
Provenance	HCLS AI Factory | Precision Biomarker Agent v1.0
Compute Platform	NVIDIA DGX Spark
Analysis Timestamp	2026-01-31T08:14:00Z
Report Version	1.0

13. PIPELINE PERFORMANCE

End-to-End Processing

Stage	Duration	Technology
Genomics (FASTQ → VCF)	120–240 min	NVIDIA Parabricks 4.6
Variant Extraction (15 loci)	< 2 min	Pipeline VCF parser
Lab Panel Ingestion	< 30 sec	Canonical data store
Biological Age Computation	< 1 min	PhenoAge + GrimAge algorithms
7-Module Analysis	< 3 min	Orchestrated pipeline
Trajectory Modeling	< 2 min	Reasoning engine
RAG Evidence Grounding	< 1 min	Med42-70B
Report Generation	< 30 sec	Template engine

Total End-to-End	< 5 hours	NVIDIA DGX Spark

Recommended Next Steps

1. **Implement priority interventions** (insulin sensitization, lipid management, methylfolate)
2. **Order Free T3 and hepatic imaging** to complete DIO2 and PNPLA3 assessment
3. **Present pharmacogenomic card** to all prescribing clinicians
4. **Schedule 3-month follow-up panel** for intervention response tracking
5. **Enroll in longitudinal monitoring** for biological age trajectory tracking

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